Siyuan Li

Personal Website: https://siyuanli.tech/ | Email: sl64343@uga.edu | WeChat: abcdefghi314159
Projects Website: https://github.com/liaoyanqing666 (250+ stars) | Tel: (+86) 13815397394 / (+1) 7063720434

Research interests: LLM reasoning, post-training, evaluation and self-evolution.

EDUCATION

University of Georgia, USA 01/2026-06/2030
Ph.D. student in Computer Science
Supervisor: **Prof. Tianming Liu**, **UGA LLM Laboratory**

Harvard University, USA 05/2026-06/2027
Visiting Ph.D. student at Harvard Medical School and Massachusetts General Hospital
Supervisor: **Prof. Xiang Li**, **CAMCA Laboratory**

Shanghai University, China 09/2021-06/2025
B.E. in Artificial Intelligence **GPA: 3.81/4.0 (93.60/100), Rank: 1/52**

SELECTED RESEARCH PROJECTS & INTERNSHIPS

Research Intern | Harvard Medical School and Massachusetts General Hospital 05/2026-Present
Advisor: Prof. Xiang Li, CAMCA Laboratory

- Developed a method for building clinical reasoning benchmarks from patient records to evaluate intervention timing, treatment decisions, and expected outcomes.
- Extracted key decision points across patients' courses of care and examined treatment-effect evidence to distinguish causal relationships from events that merely followed treatment.
- Designed longitudinal evaluations of how LLMs use earlier patient information to make later clinical decisions and predict treatment outcomes.
- Collaborated with clinicians on a review framework for benchmark design and causal annotations, supporting data preparation for future work on medical world models.

Railbound Benchmark | Visual Reasoning and Planning 10/2026-Present

- Built a track-laying puzzle benchmark with limited resources, synchronized trains, arrival-order constraints, and interacting mechanisms to assess spatial understanding, temporal reasoning, and multi-step planning.
- Constructed a deterministic simulator with board images, structured states, and reference solutions; executed proposed plans and returned feedback on constraint violations, collisions, and loops.
- Designed comparisons of images with rules, images without rules, and text grids with rules to examine the roles of visual perception, mechanism understanding, and planning.
- Evaluating multiple models through direct text or image reasoning without tool calls or code execution, including their ability to correct plans across multiple turns.

Research Intern | Guangming Laboratory | LLM Post-training 09/2025-01/2026

- Constructed a high-quality medical Instruction Tuning dataset by performing in-depth cleaning and structuring of large-scale private clinical records, and designing diverse QA templates to ensure comprehensive data coverage.
- Developed a fully automated Supervised Fine-Tuning (SFT) pipeline, streamlining the end-to-end workflow from heterogeneous data loading and dynamic prompt construction to efficient model training.
- Generated answer-conditioned Chain-of-Thought (CoT) reasoning traces from correct answers and incorporated these rationales into supervised fine-tuning for medical reasoning.
- Applied Reinforcement Learning for subsequent policy optimization in clinical decision-making, with a focus on multi-step medical reasoning.

Multimodal Retrieval-Augmented Generation and Medical VLM Fine-Tuning | Graduation Project 05/2025

- Designed and implemented a medical visual question answering system combining Retrieval-Augmented Generation (RAG) and multimodal large language model (MLLM) fine-tuning.

- Developed a multimodal RAG framework supporting joint image-text input, featuring multiple retrieval paradigms such as joint embedding, label-guided retrieval, and image-text pair binding.
- Fine-tuned the Qwen2.5-VL model in two stages with LoRA using Chinese medical QA and image-text datasets, yielding the Qwen2.5-VL-Med model with domain-specific reasoning capabilities.
- Built a modular web-based interactive system supporting local/cloud API deployment, multimodal input, streaming response, and history tracking.
- Compared the base model, retrieval-augmented model, and retrieval-plus-fine-tuning system on OmniMedVQA, and evaluated alternative retrieval strategies to examine their separate contributions.

Intern at Huawei | Department of Ascend Computing Inference Development | AI (NLP) engineer 12/2024-03/2025

Location: Huawei Lianqiu Lake R&D Center, Shanghai

- Reported and fixed a bug (issue #11978 and PR #11979), with the fix merged into vLLM's main branch.
- Contributed to porting vLLM to the Ascend NPU platform (vllm-ascend), with responsibilities including unit testing, operator adaptation, and speculative decoding support.

Research Intern at Westlake University | Rule Discovery in Physical Data/Video 07/2023-06/2024

Advisor: Prof. Tailin Wu (Westlake University), Prof. Sebastian Musslick (Brown University)

- Developed a Transformer-based symbolic regression model and generated syntactically valid equations in prefix notation for training and controlled experiments.
- Created synthetic videos of objects moving according to known formulas against randomized backgrounds, and built a multi-component autoencoder to separate motion and appearance representations.
- Represented motion with two latent scalars and appearance with a separate code; predicted subsequent motion states and decoded them to reconstruct the next frame.
- Combined the motion encoder and symbolic regressor to investigate equation discovery from video; mapping learned latent variables to interpretable physical quantities and valid equations remains an open challenge.

PUBLICATIONS AND PREPRINTS (FIRST/CO-FIRST AUTHORED)

Shengyang Zhong†, **Siyuan Li**†, et al. *Imaging-Centered Artificial Intelligence for Precision Cancer Immunotherapy: Multimodal Biomarkers, Treatment-Effect Estimation, and Clinical Translation.* Research Square preprint, 2026. DOI: 10.21203/rs.3.rs-10468208/v1. Under review at *Meta-Radiology*.

Siyuan Li, et al. *ASTELD: A Six-Axis Classification Framework for Autonomous AI Agents - Design, Evaluation, and an OpenClaw Case Study.* arXiv preprint, 2026. arXiv:2608.05201.

Siyuan Li, et al. *The Study of Human Preference Based on Integrated Analysis of N1 and LPP Components.* arXiv preprint, 2025. arXiv:2505.19879.

Yuhang Guo†, **Siyuan Li**†, Jinxuan Wu†. *Research advanced in offline handwritten signature verification. Applied and Computational Engineering*, 2023, 6(1): 1236–1244. DOI: 10.54254/2755-2721/6/20230653.

† Equal contribution.

SKILLS

Engineering: Python, C++, JavaScript, HTML / CSS, Git / GitHub, Linux and RemoteSSH, LaTeX

LLM/MLLM post-training: Supervised fine-tuning, LoRA, reasoning-trace generation, medical model adaptation

Evaluation & multimodal learning: Benchmark design, visual planning, multimodal retrieval, cross-modal alignment

Agents & automation: LLM-assisted research, vibe-coding, Codex computer-use, content publishing workflows

Frameworks & Inference: PyTorch, Hugging Face Transformers, vLLM

AWARDS (SELECTED)

ASC Student Supercomputer Challenge **National Second Prize** 02/2024

ICPC (International Collegiate Programming Contest) Asia Regional Contest (Hefei) **Bronze Medal** 11/2023

ICPC (International Collegiate Programming Contest) Asia Regional Contest (Nanjing) **Bronze Medal** 11/2022

Group A of C/C++ Division of Blue Bridge Programming Cup **National Third Prize** 06/2023

CCPC (China Collegiate Programming Contest) Shanghai Programming Contest **Silver Medal** 10/2022